A Study on Transformer-Based Multimodal Learning for Pulmonary Disease Diagnosis

Abstract

In clinical practice, physicians rarely rely on a single source of information when diagnosing pulmonary diseases. Instead, diagnostic decisions are formed by combining chest radiographs with laboratory findings, vital signs, patient history, and other structured clinical parameters. Despite this, many existing artificial intelligence systems continue to operate on a single modality, most commonly medical imaging, limiting their ability to reflect real-world clinical reasoning. Although these models have demonstrated promising diagnostic performance, their inability to effectively integrate complementary patient information restricts both their robustness and clinical applicability, particularly in complex healthcare environments where multiple sources of evidence are routinely considered.

The emergence of transformer-based multimodal learning provides a promising direction for addressing this limitation by enabling medical images and structured clinical information to be processed within a unified framework. Through attention-based learning, these models can capture meaningful relationships between radiographic findings and patient-specific clinical parameters, allowing diagnostic decisions to be informed by multiple complementary modalities rather than by imaging alone. Furthermore, recent advances have shown that multimodal learning can improve diagnostic performance, remain robust in the presence of incomplete clinical information, provide interpretable attention-based visualisations, and support validation across diverse clinical datasets, demonstrating its growing potential for real-world healthcare applications.

This seminar examines recent developments in transformer-based multimodal learning for pulmonary disease diagnosis, with emphasis on integrating chest radiographs and structured clinical data into a single diagnostic framework. It reviews how multimodal architectures, attention-based feature fusion, and complementary clinical information collectively contribute to more reliable and clinically meaningful diagnostic systems, while also discussing recent progress in multimodal representation learning, fusion strategies, longitudinal disease modelling, and large-scale clinical validation. Overall, the study highlights transformer-based multimodal learning as a significant step toward more accurate, interpretable, and clinically relevant artificial intelligence systems for pulmonary disease diagnosis.

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